迴歸預測

Regression & Forecasting

by

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The Role of Forecasting

- Forecasting is a vital function and affects every significant management decision.
  - ✓ Finance and accounting use forecasts as the basis for budgeting and cost control.
  - ✓ Sales and marketing relies on forecasts to make key decisions such as new product planning and personnel compensation.
  - ✓ Production uses forecasts to select suppliers; determine capacity requirements; and drive decisions about purchasing, staffing, and inventory.

預測的種類

- There are four basic types of forecasts.
  - Qualitative (老闆、主管或專家說了算)
  - Time series analysis (根據過去歷史資料)
  - Causal relationships (迴歸就是用 X預測Y)
  - Simulation (需考慮機率或不確定性)

- Time series (時間序列) analysis is based on the idea that data relating to past demand can be used to predict future demand.

Time Series Analysis

- Using the past to predict the future

Short term – forecasting less than 3 months

- Used mainly for tactical decisions

Medium term – forecasting 3 months to 2 years

- Used to develop a strategy that will be implemented over the next 6 to 18 months (e.g., meeting demand)

Long term – forecasting greater than 2 years

- Useful for detecting general trends and identifying major turning points

Forecasting Method Selection Guide

Forecasting Method	Amount of Historical Data	Data Pattern	Forecast Horizon
Simple moving average	6 to 12 months; weekly data are often used	Stationary (i.e., no trend or seasonality)	Short
Weighted moving average and simple exponential smoothing	5 to 10 observations needed to start	Stationary	Short
Exponential smoothing with trend	5 to 10 observations needed to start	Stationary and trend	Short
Linear regression	10 to 20 observations	Stationary, trend, and seasonality	Short to medium

迴歸績效指標衡量

$$MSE = \frac{\sum(\text{Actual}_t - \text{Forecast}_t)^2}{n}$$

Mean Squared Error

$$MAD = \frac{\sum|\text{Actual}_t - \text{Forecast}_t|}{n}$$

Mean Absolute Deviation

$$MAPE = \frac{\sum\left|\frac{\text{Actual}_t - \text{Forecast}_t}{\text{Actual}_t}\right|}{n} * 100\%$$

Mean Absolute Percentage Error

Tracking Signals

- A measure that indicates whether a method of forecasting is accurately predicting actual changes in demand
- Useful when forecast systems are computerized because it alerts analysts when forecast are getting far from desirable limits (CFE: cumulative forecasting error)

$$\text{Tracking signal} = \frac{\text{CFE}}{\text{MAD}}$$

- Each period, the CFE and MAD are updated to reflect current error, and the tracking signal is compared to some predetermined limits

Tracking Signals

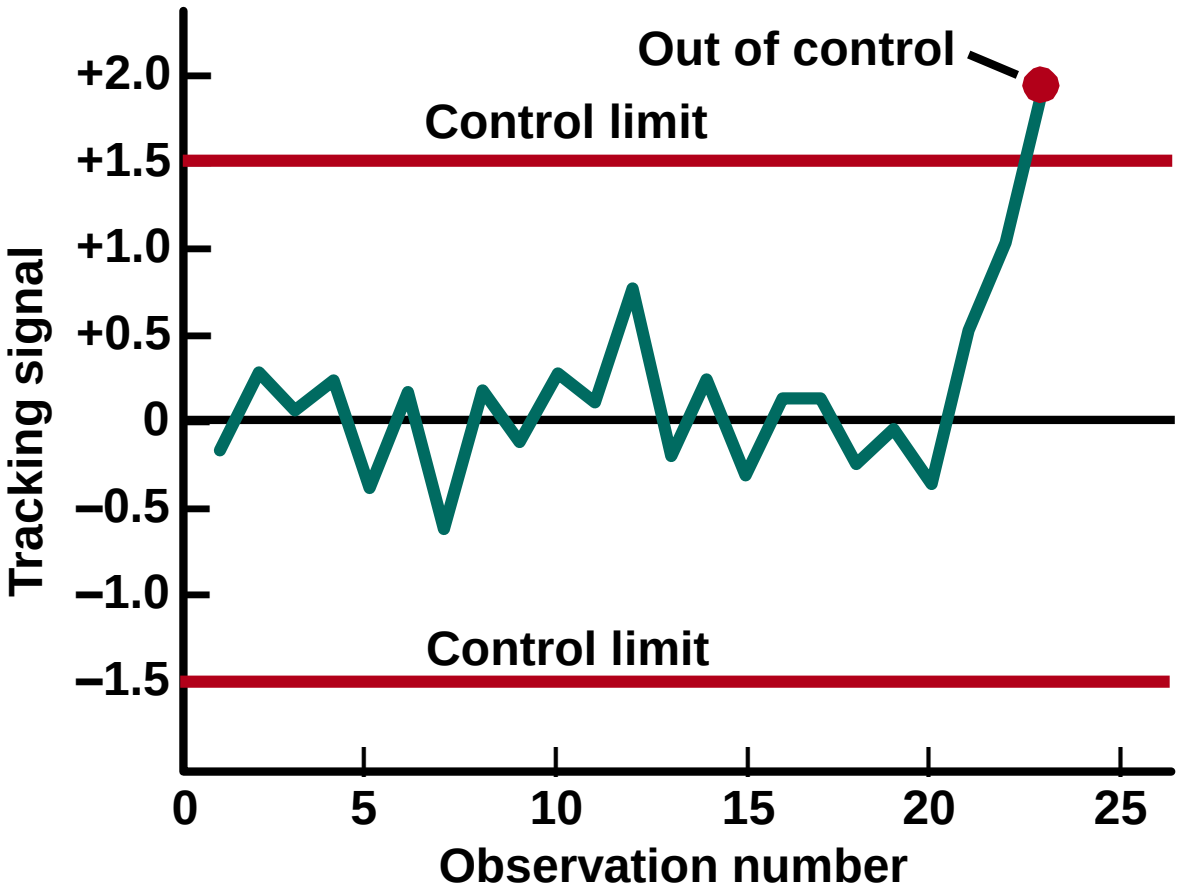


Figure 13.7 – Tracking Signal

Simple Moving Average

- Forecast is the average of a fixed number of past periods (以過去歷史資料的平均預測未來).
- Useful when demand is not growing or declining rapidly and no seasonality is present.
- Removes some of the random fluctuation from the data.
- Selecting the period length is important.
  - Longer periods provide more smoothing.
  - Shorter periods react to trends more quickly.

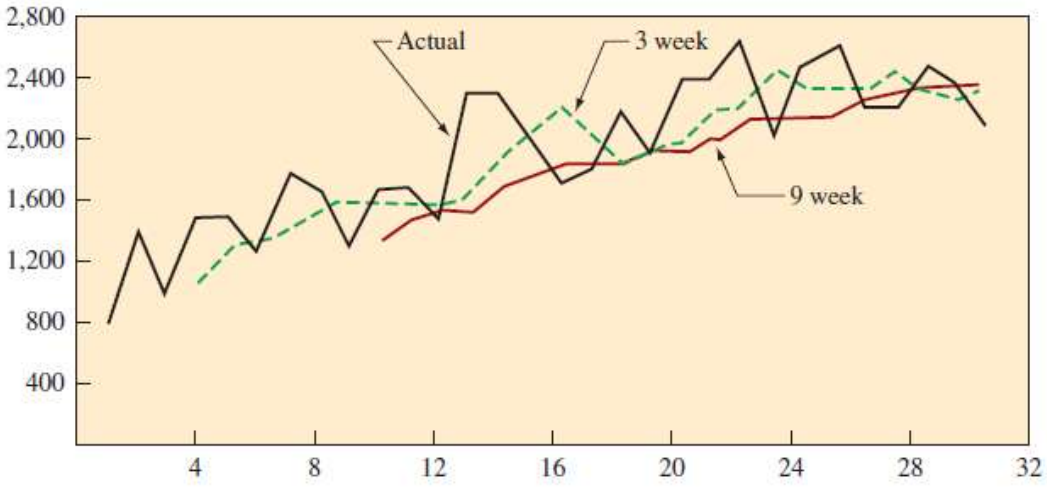
Simple Moving Average Formula

就是股市當中的五日、十日、20日移動平均線

- $F_t = \frac{A_{t-1}+A_{t-2}+A_{t-3}+\cdots+A_{t-n}}{n}$
- F_t - Forecast for period t (the next period)
- n – Number of periods to be averaged
- A_{t-1} - Actual occurrence in the past period
- A_{t-2}, A_{t-3}, and A_{t-n} - Actual occurrences two periods ago, three periods ago, and so on up to n periods ago

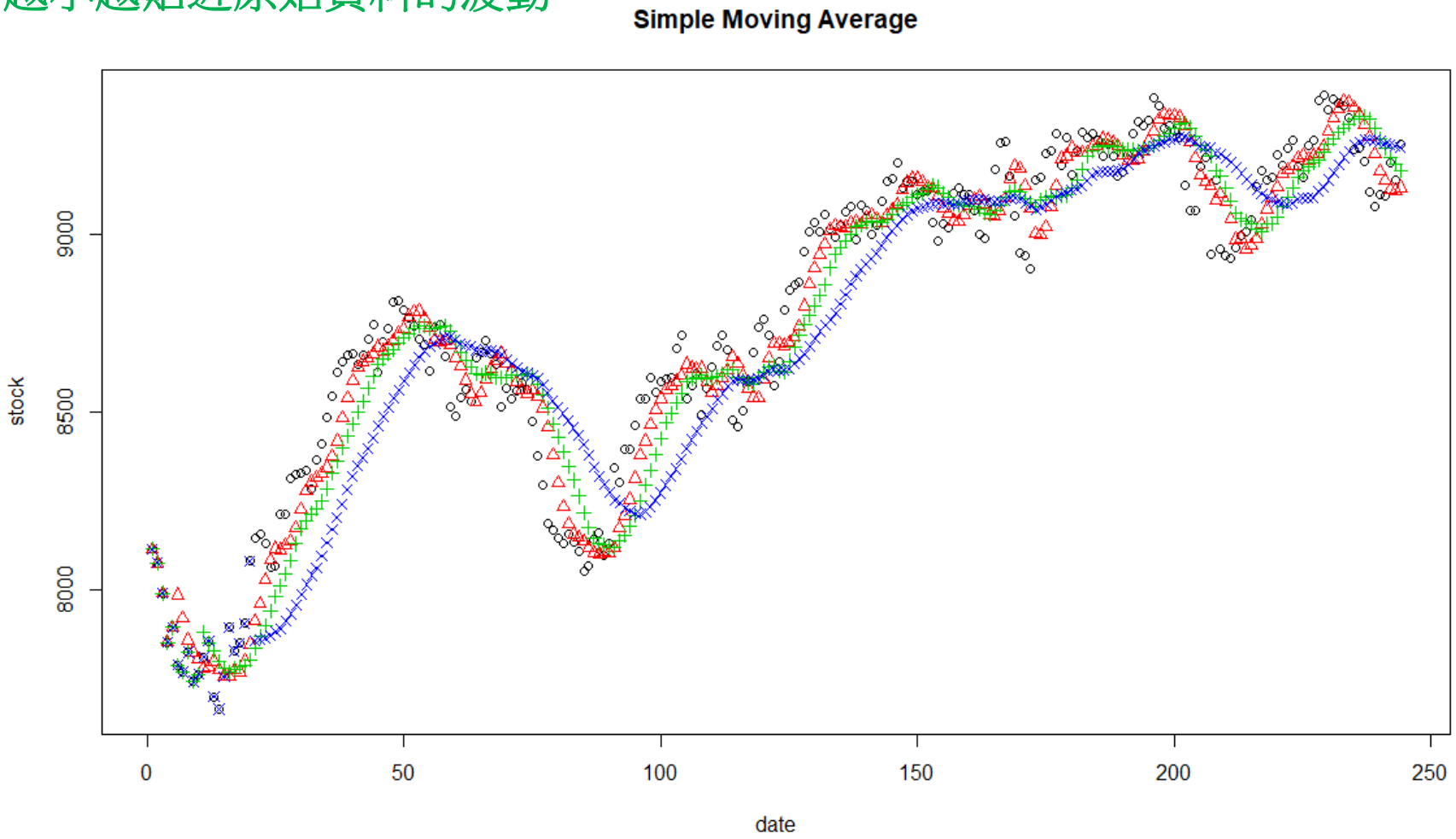
Simple Moving Average – Example

Week	Demand	3 Week	9 Week	Week	Demand	3 Week	9 Week
1	800			16	1,700	2,200	1,811
2	1,400			17	1,800	2,000	1,800
3	1,000			18	2,200	1,833	1,811
4	1,500	1,067		19	1,900	1,900	1,911
5	1,500	1,300		20	2,400	1,967	1,933
6	1,300	1,333		21	2,400	2,167	2,011
7	1,800	1,433		22	2,600	2,233	2,111
8	1,700	1,533		23	2,000	2,467	2,144
9	1,300	1,600		24	2,500	2,333	2,111
10	1,700	1,600	1,367	25	2,600	2,367	2,167
11	1,700	1,567	1,467	26	2,200	2,367	2,267
12	1,500	1,567	1,500	27	2,200	2,433	2,311
13	2,300	1,633	1,556	28	2,500	2,333	2,311
14	2,300	1,833	1,644	29	2,400	2,300	2,378
			1,733	30	2,100	2,367	2,378



Simple Moving Average – 2016 stock

*n*越小越貼近原始資料的波動



Weighted Moving Average

- The simple moving average formula implies equal weighting for all periods (簡單移動平均假設不同時間的資料權重均相同).
- A weighted moving average allows unequal weighting of prior time periods.
  - The sum of the weights must be equal to one.
  - Often, more recent periods are given higher weights than periods farther in the past.

$$F_t = w_1 A_{t-1} + w_2 A_{t-2} + \cdots + w_n A_{t-n}$$

  - 加權移動平均的問題在於不知如何決定不同時間點的資料權重

Exponential Smoothing

- A weighted average method that includes all past data in the forecasting calculation
- More recent results weighted more heavily
- The most used of all forecasting techniques
- An integral part of computerized forecasting
- Well accepted for six reasons
  - ✓ Exponential models are surprisingly accurate.
  - ✓ Formulating an exponential model is relatively easy.
  - ✓ Little computation is required to use the model.
  - ✓ Computer storage requirements are small.
  - ✓ Tests for accuracy are easy to compute.

Exponential Smoothing Model

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1}) \quad = \alpha A_{t-1} + (1-\alpha) F_{t-1}$$

$$= \alpha A_{t-1} + (1-\alpha)[\alpha A_{t-2} + (1-\alpha) F_{t-2}] \ldots\ldots$$

where

Alpha一般介於0.1~0.5之間

F_t = The exponentially smoothed forecast for period t

F_{t-1} = The exponentially smoothed forecast made for the prior period

A_{t-1} = The actual demand in the prior period

α = The desired response rate, or smoothing constant

指數平滑法假設越近的資料越重要，越遠的資料越不重要

Exponential Smoothing Example

Week	Demand	Forecast
1	820	820
2	775	820
3	680	811
4	655	785
5	750	759
6	802	757
7	798	766
8	689	772
9	775	756
10		760

$F_2 = F_1 + \alpha(A_1 - F_1)$ =820 + 0.2(820-820)

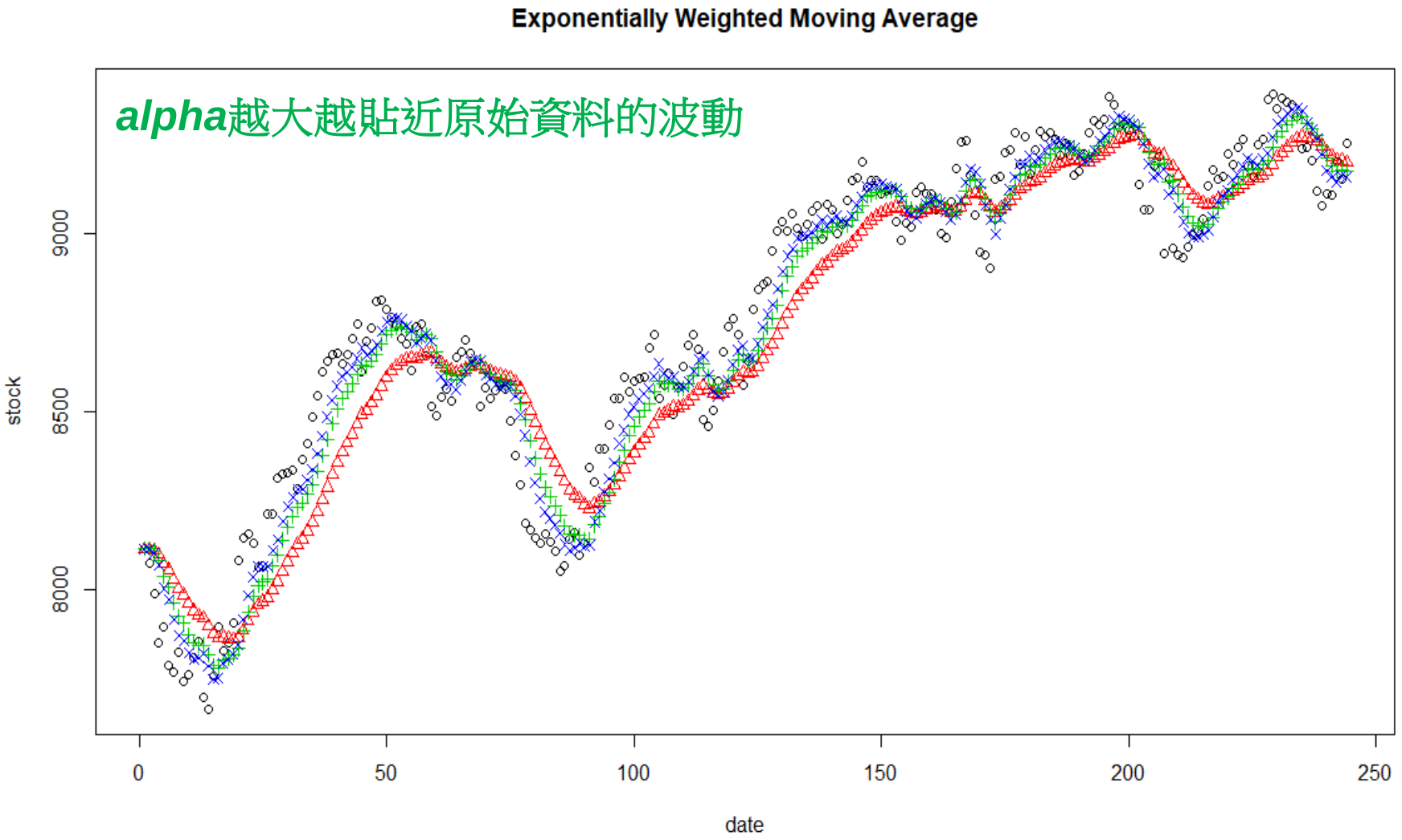
$F_4 = F_3 + \alpha(A_3 - F_3)$ =811+ 0.2(680-811)

$F_6 = F_5 + \alpha(A_5 - F_5)$ =759+ 0.2(750-759)

$F_8 = F_7 + \alpha(A_7 - F_7)$ =766+ 0.2(798-766)

$F_{10} = F_9 + \alpha(A_9 - F_9)$ =756+ 0.2(775-756)

Exponential Smoothing– 2016 stock



Criteria for Selecting Methods

- Criteria to use in making forecast method and parameter choices include
  1. Minimizing MAPE, MAD, or MSE
  2. Meeting managerial expectations of changes in the components of demand
  3. Minimizing the forecast error last period

- Statistical performance measures can be used
  1. For projections of more stable demand patterns, use lower α values or larger n values
  2. For projections of more dynamic demand patterns try higher α values or smaller n values

Holt-Winter's moving average

$$L_t = \alpha(y_t - S_{t-s}) + (1-\alpha)(L_{t-1} + b_{t-1})$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

$$S_t = \gamma(y_t - L_t) + (1-\gamma)S_{t-s}$$

$$F_{t+k} = L_t + kb_t + S_{t+k-s}$$

Forecast consists of level (L), trend (T), and seasonal (S) components

MLR的判定係數

傳統上，MLR (multiple linear regression)迴歸模型的評估常使用判定係數(coefficient of determination): $Y=a_0+a_1x_1+a_2x_2+a_3x_3+\ldots$

$$R^2 = SSR/SST$$

或調整後判定係數（adjusted coefficient of determination:）避免使用過多沒必要的X (k表示X的個數, T表示樣本數)

$$\bar{R}^2 = 1-\frac{T-1}{T-k}(1-R^2)$$

MARS介紹

Multivariate Adaptive Regression Splines
(MARS多元自適雲形迴歸)

統計物理學家 Friedman 於 1991 年提出多元適應雲形迴歸，其屬於一種多變量無母數迴歸處理的程序。其能夠正確的預測連續與二元的相依變數，找出隱藏在高維度資料中的複雜結構及變數的交互作用。

過去對於資料之分析，常常不論資料是否符合常態，均用線性迴歸進行實證，導致模型適用性出現問題，而MARS為多變量無母數分析技術，相較於傳統之線性迴歸，可解決資料非線性、非常態等問題。

MARS原理

MARS 基本概念為運用分段基底函數(Spline Basis Function, BF)加總，以組合較有彈性的預測模型解決複雜的資料配適。

$$\hat{f}(x) = a_0 + \sum_{m=1}^{M} a_m \prod_{k=1}^{K_m} [s_{km} \cdot (x_{v(k,m)} - t_{km})]_+$$

有關BF之後段累程之部分，主要是根據需求變化而得：

$$B_m(x) = \prod_{k=1}^{K_m} H[s_{km} \cdot (x_{v(k,m)} - t_{km})]_+$$

上式中a_0與a_m皆為參數值；M為BF的個數； K_m為切割之節點(knot)個數； S_{km}之值為+1或-1，作用為顯示方向；v(k,m)是對變數的標示；t_{km}是各節點的臨界值。

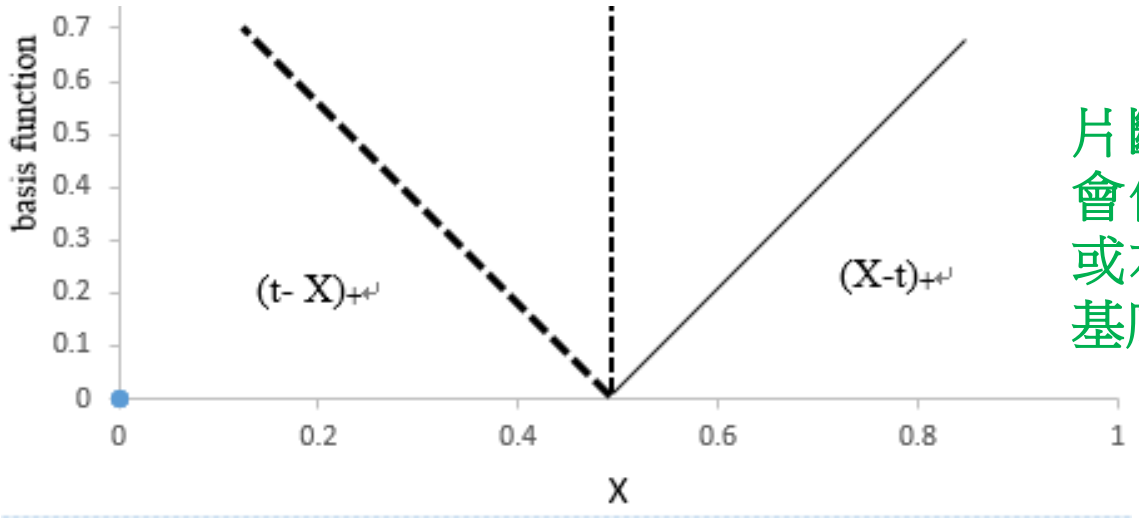
MARS基底函數

- 模式中所以用的片段線性基底函數形式如下(+表示正值)：

$(X-t)_+ = \{X-t, \text{ if } X > t \text{ ; } 0, \text{ otherwise}\}$，

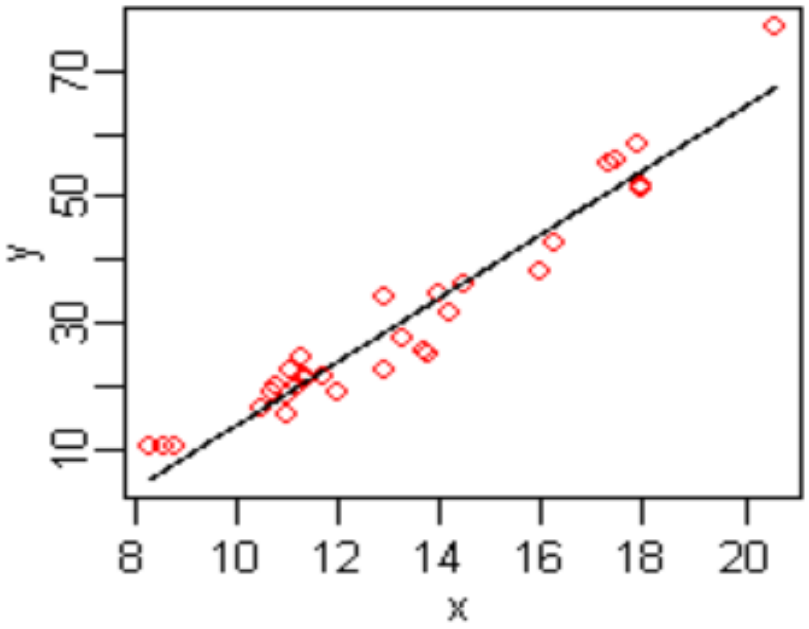
$(t-X)_+ = \{t-X, \text{ if } X < t \text{ ; } 0, \text{ otherwise}\}$

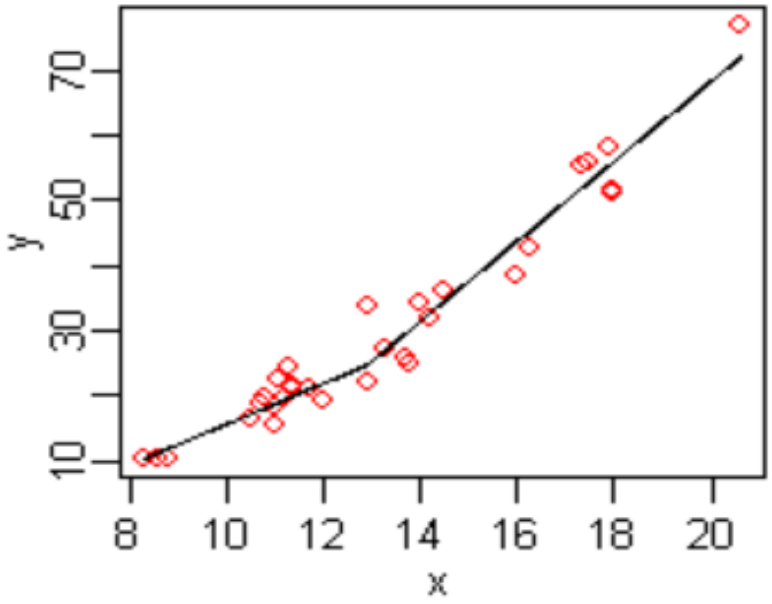
- X為預測變數，t為獨立的觀察值，表示折點(knot)的位置；通常$(X-t)_+$與$(t-X)_+$會成對出現。例如：t=0.5時，基底函數為$(X-0.5)_+$與$(0.5-X)_+$，如圖二所示。



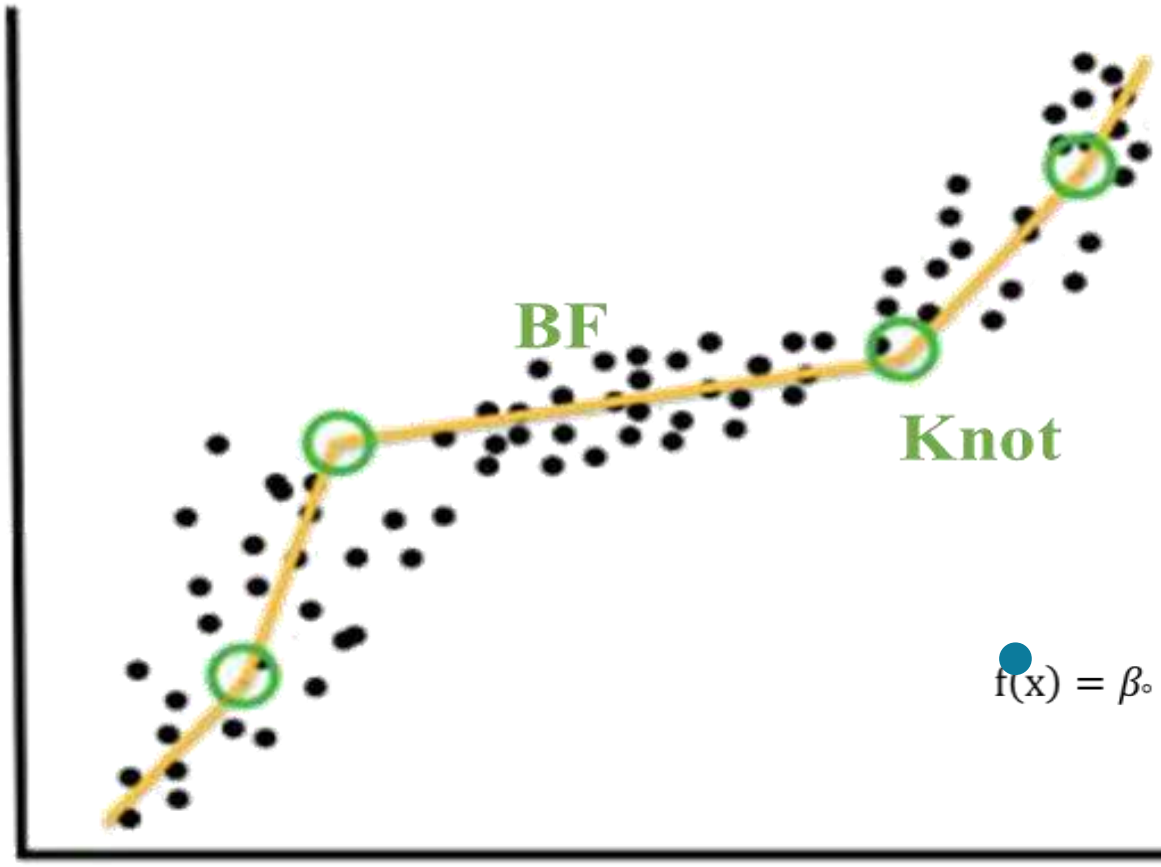
片斷線性基底函數，只會依據測試，選取左邊或右邊的任一片段當作基底的一員

MARS與MLR的比較





MARS使用片段線性基底進行估計



$$f(x) = \beta_0 + \sum_{m=1}^{M} \beta_m h_m(X)$$

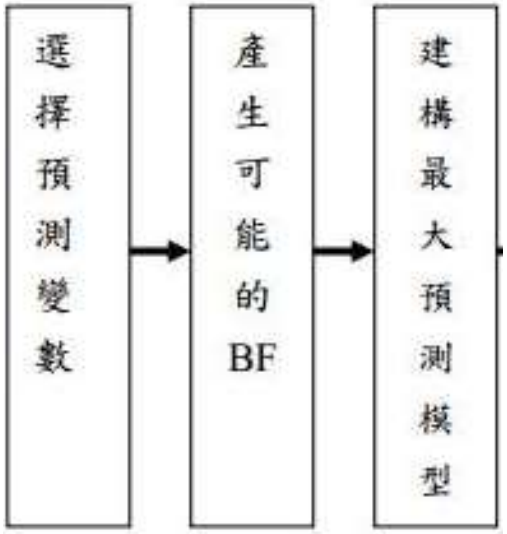
MARS執行過程

篩選的過程採用前推式與後推式(Forward and Backward Stepwise Procedure) 的演算法：

(一)前推式階段

不斷增加解釋方程式(所有的主要效果、新的主要影響因子、折點或交互作用)的個數，使模型不斷成長直到找出一個完整模型為止。

注意：在此階段會有過度配適(over fitting)的問題



MARS執行過程

(二)後推式階段 (general cross validation)

此階段根據缺適性的概念加以判斷，選擇重要變數的過程中使用 GCV 法為判定原則，所以當一變數由模型中去除後使 GCV 顯著降低，則可知此變數為重要關鍵要素。

$$GCV(M) = \frac{1}{N}\sum_{i=1}^{N}\frac{\left[y_i - \hat{f}_M(\pi_i)\right]^2}{\left[1-\frac{C(M)}{N}\right]^2}$$

上式中分子為實際值減預測值，
C(M)為採用M個BF所花費的成本。



什麼時候需要偏差迴歸?

- 脊迴歸 (Ridge Regression)
- 拉索迴歸(Lasso Regression)
- 彈性網路(Elastic Net)
- 可適性拉索迴歸(Adaptive Lasso Regression)

變數很多，但資料量不夠時: 如30個機台參數，但只有50筆資料

脊迴歸 (Ridge Regression)

- **進行迴歸分析時，當遭遇以下幾種情況時，可能要注意迴歸模式中是否存在多元共線性(multi-collinearity)的問題：**
  - ✓ 共線性指標超過標準
  - ✓ 迴歸係數的方向性與相關係數相反
  - ✓ 解釋力*R*平方很高，但個別變項的係數未達顯著水準

- 解決共線性的方法如下：
  - ✓ 刪除共線性X變項
  - ✓ 利用主成分分析將線性重合的自變數重新建構成新的潛在變項，來替代原有的自變數
  - ✓ 使用「偏差迴歸分析法」(Biased Regression Analysis,簡稱 BRA)，如使用 Ridge regression analysis (脊迴歸)。

脊迴歸 (Ridge)

- 在 LS（least square）最小平方法估計法下：β具有 BLUE（best linear unbiased estimator）特性，即不偏性和有效性

$$\hat{\beta} = \arg\min_{\beta}(Y - X\beta)^T(Y - X\beta) = (X^TX)^{-1}X^TY$$

- 而Ridge regression則是犧牲一點不偏性，但仍具有效性，模型如下：

誤差

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta}\{\|Y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\}$$

$$Y = (y_1, \cdots, y_n)^T, \qquad n \times 1$$

$$X = (X_1, \cdots, X_p), \qquad n \times p$$

Penalty

套索迴歸 (Lasso)

套索迴歸（Least Absolute Shrinkage and Selection Operator- Lasso）

- 首先由TIBSHIRAN提出，是處理多重共線性問題的迴歸方法
- 它是一種壓縮估計，基本思想是在迴歸係數的絕對值之和小於一個常數的約束條件下，使殘差平方和最小化，從而能夠產生某些嚴格等於0的迴歸係數(刪除變數)，得到可以解釋的模型

誤差

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \{ \|Y - X\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

Penalty

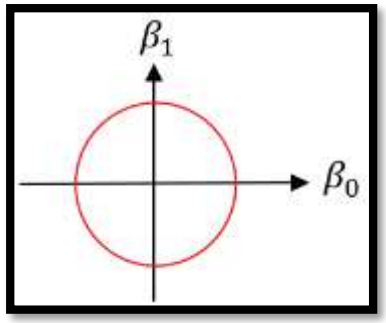
脊迴歸 vs. 套索迴歸

向量範數（vector norm）

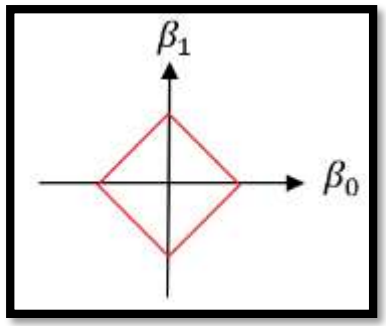
- 是具有「長度」概念的函數，其為向量空間內的所有向量賦予非零的正長度或大小。

$$\vec{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

✓ **L2-norm** = $\|\beta\|_2 = \sqrt{\beta_0^2 + \beta_1^2}$ → *脊迴歸*

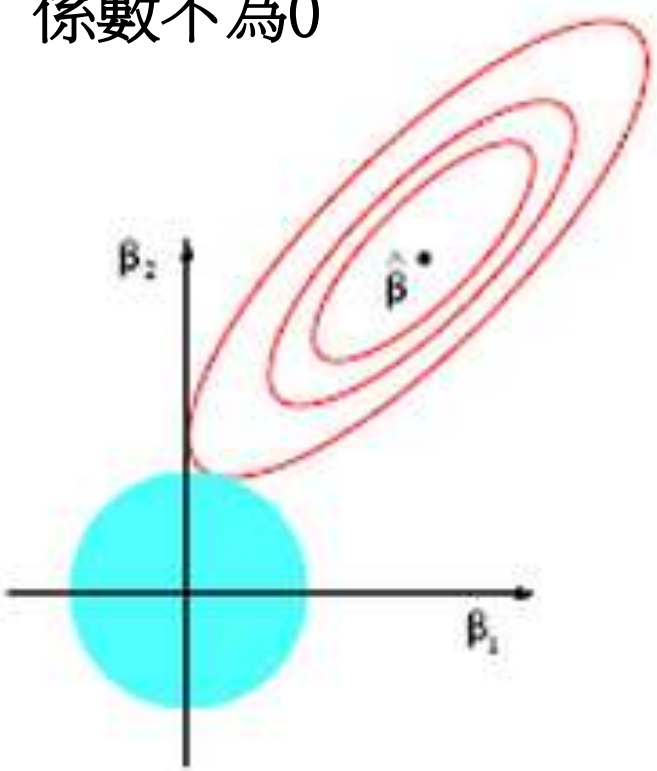


✓ **L1-norm** = $\|\beta\|_1 = |\beta_0| + |\beta_1|$ → *套索迴歸*

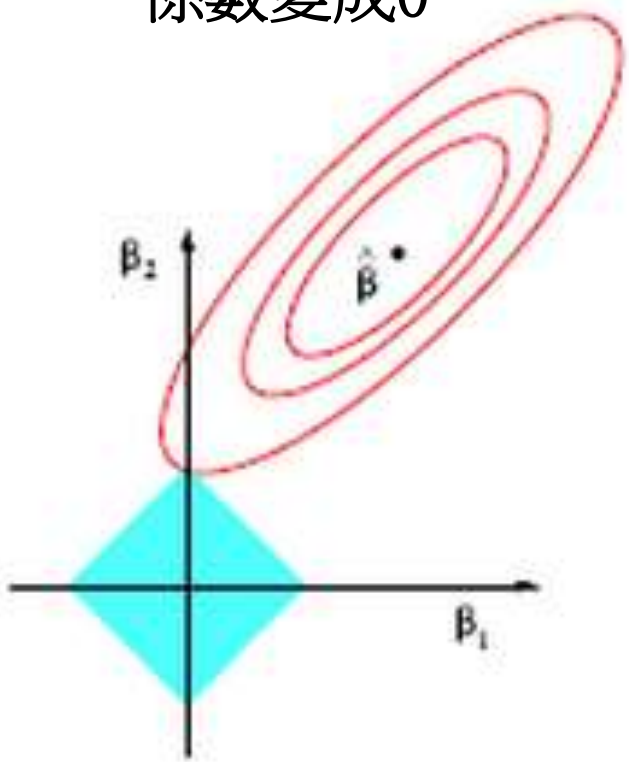


脊迴歸 vs. 套索迴歸

脊迴歸：
係數不為0



套索迴歸：
係數變成0

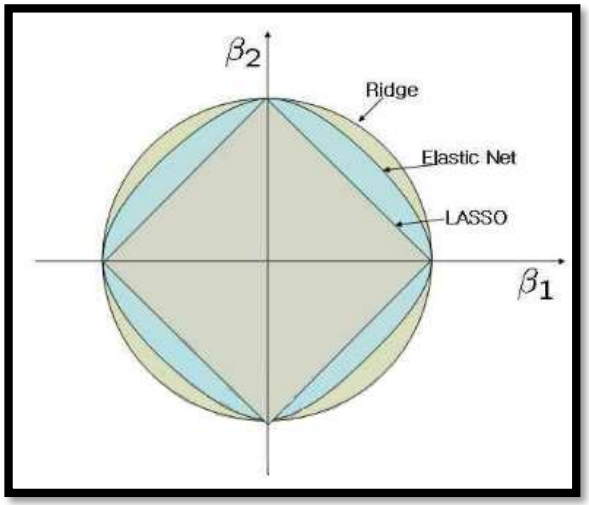


彈性網路 (ElasticNet)

$$\hat{\beta}^{enet} = \arg\min_{\beta} \{(Y - X\beta)^T (Y - X\beta) + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2\}$$

↑**Penalty**

- Elastic net融合了norm1和norm2兩種方法



- ElasticNet對於*p*遠大於n或者嚴重的多重共線性情況有明顯的效果
- 一般來說，彈性網路是脊迴歸和拉索迴歸很好的折衷

自適性套索迴歸

自適性套索迴歸（Adaptive LASSO, ALSSO）

$$\hat{\beta} = \arg\min_{\beta} \left\| \mathbf{y} - \sum_{j=1}^{p} \mathbf{x}_j \beta_j \right\|^2 + \lambda \sum_{j=1}^{p} w_j |\beta_j|$$

$$w_i = \frac{1}{(\hat{\beta}_i)^{\gamma}}.$$

Penalty

特殊迴歸的綜合比較

- Ridge Regression：
  - Good for multi-collinearity, group selection
  - Not good for variable selection.

- Lasso Regression：
  - Good for variable selection
  - Not good for group selection for strongly correlated predictors.

- Elastic Net：
  - Combining the strength of Ridge regression with Lasso regression

- Adaptive Lasso：
  - Enhancing Lasso regression to focus on variable selection

機器學習的非線性迴歸

- 最近鄰居法 (KNN- *K* nearest neighbor)
- 迴歸樹 (RT- Regression Tree/CART)
- 隨機森林 (RF- Random Forest)
- 集成式演算 (AB-Adaboost/GB-Gradient boosting)
- 倒傳遞網路 (BPN- Back Propagation Network)
- 支援向量法 (SVR- Support Vector Regression)

迴歸演算法總結

- 迴歸與分類的差別在於因(Y)變數是連續數值或間斷類別
- 除SVR、BPN、KNN(只接受數值型X變數)外，其他迴歸幾乎都能提供X變數對預測Y的重要性判斷
- RT(CART)、RF較不易受離群值影響，且能計算*X*變數的重要性(MLR, MARS易受離群值影響)
- 線性迴歸的解釋力(R^2)不佳，不表示X對Y沒有預測力，僅表示該份資料不適用線性模型來處理
- Ridge及Lasso適合處理X變數特多但資料筆數不夠的狀況 (新產品或新製程剛開發出來時)

平衡計分卡四大構面

以24個因子X預測
EPS (每股盈餘)- Y

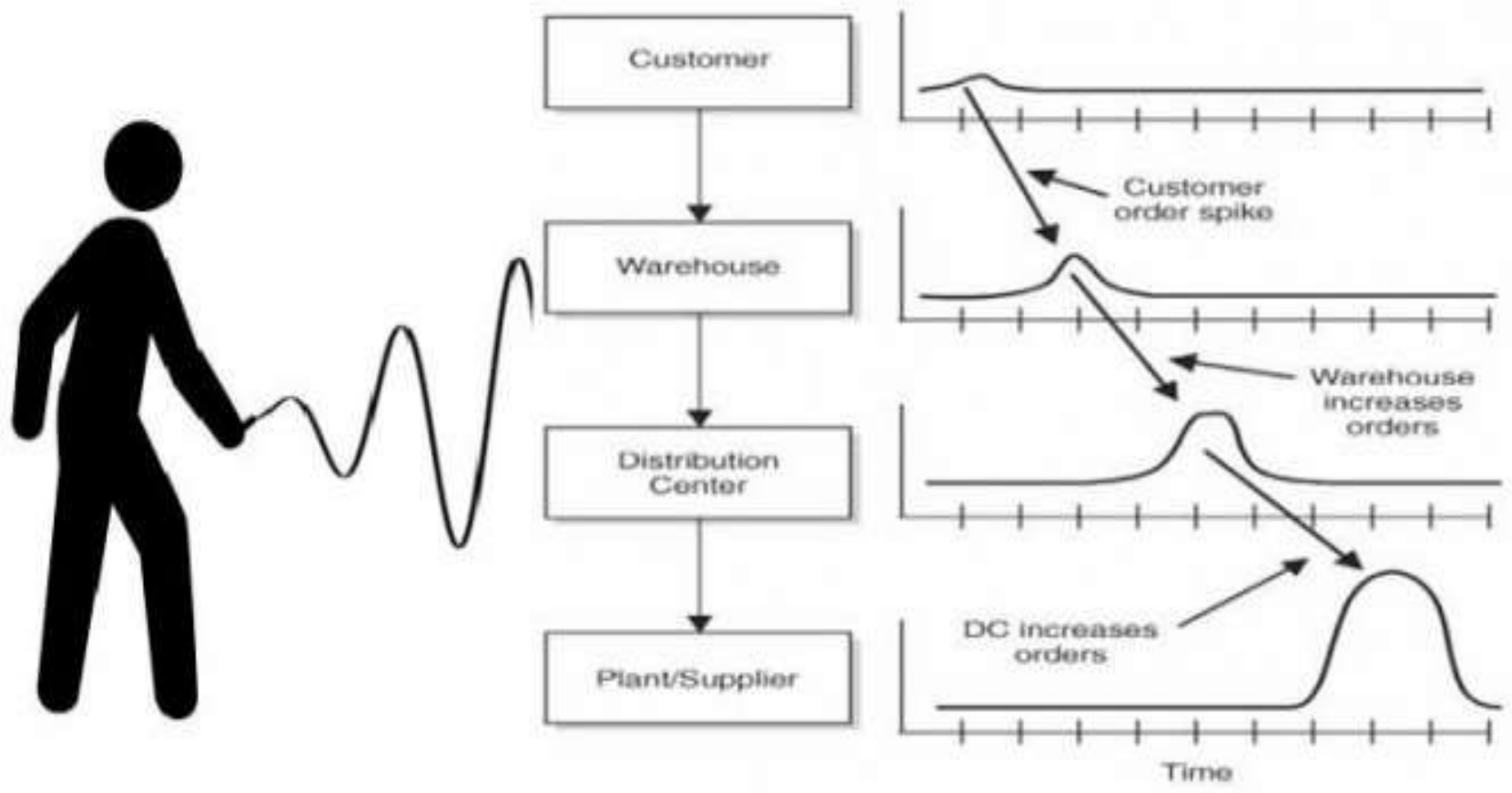
A1	Debt ratio
A2	Current ratio
A3	Quick ratio
A4	Turnover rate of accounts receivable
A5	Turnover rate of inventory
A6	Turnover rate of fixed assets
A7	Ratio of cash flow
A8	Product sale revenue
A9	Turnover rate of total assets
A10	Days of inventory turn
B1	Number of employees
B2	Turnover rate of employees
B3	Average working years of employees
B4	Average age of employees
B5	Patent term
C1	Product varieties
C2	Profit margin of major products
C3	Proportion of major products
C4	Export ratio of major products
D1	Ratio of employees' stock bonus
D2	Ratio of R&D investment
D3	Ratio of operational expense
D4	Growth rate of revenues
D5	Operational efficiency

Overall Comparison: Ridge, Lasso, ElasNet

MSE Error
Ridge: 1.644,
Lasso: 1.782,
ElasNet: 1.701

Intercept	-1.971	-0.725	0.658
A1	-1.003		
A2	-1.343		
A3	0.506		
A4	-0.648		
A5	-0.389		
A6	1.112		
A7	1.336	2.750	0.829
A8	4.072		
A9	2.668	2.886	1.898
A10	-1.336	-2.751	-0.681
B1	-1.252		
B2	0.483		
B3	1.407		-0.197
B4	0.232		
B5	4.190		
C1	-0.730	-1.352	-0.066
C2	0.300		
C3	0.953		
C4	-0.642		
D1	2.607		
D2	3.959		
D3	-3.837	-1.617	-1.416
D4	0.159	1.873	0.515
D5	0.910	1.330	0.874

Supply chain analytics



Supply chain forecasting

Y=f(x1, x2, x3, …., xn), Y is a dependent variable, xi are predictors
(先定義X、Y變數)

Y can be production capacity, unit prices, sales revenues, sales volume (shipments), ??

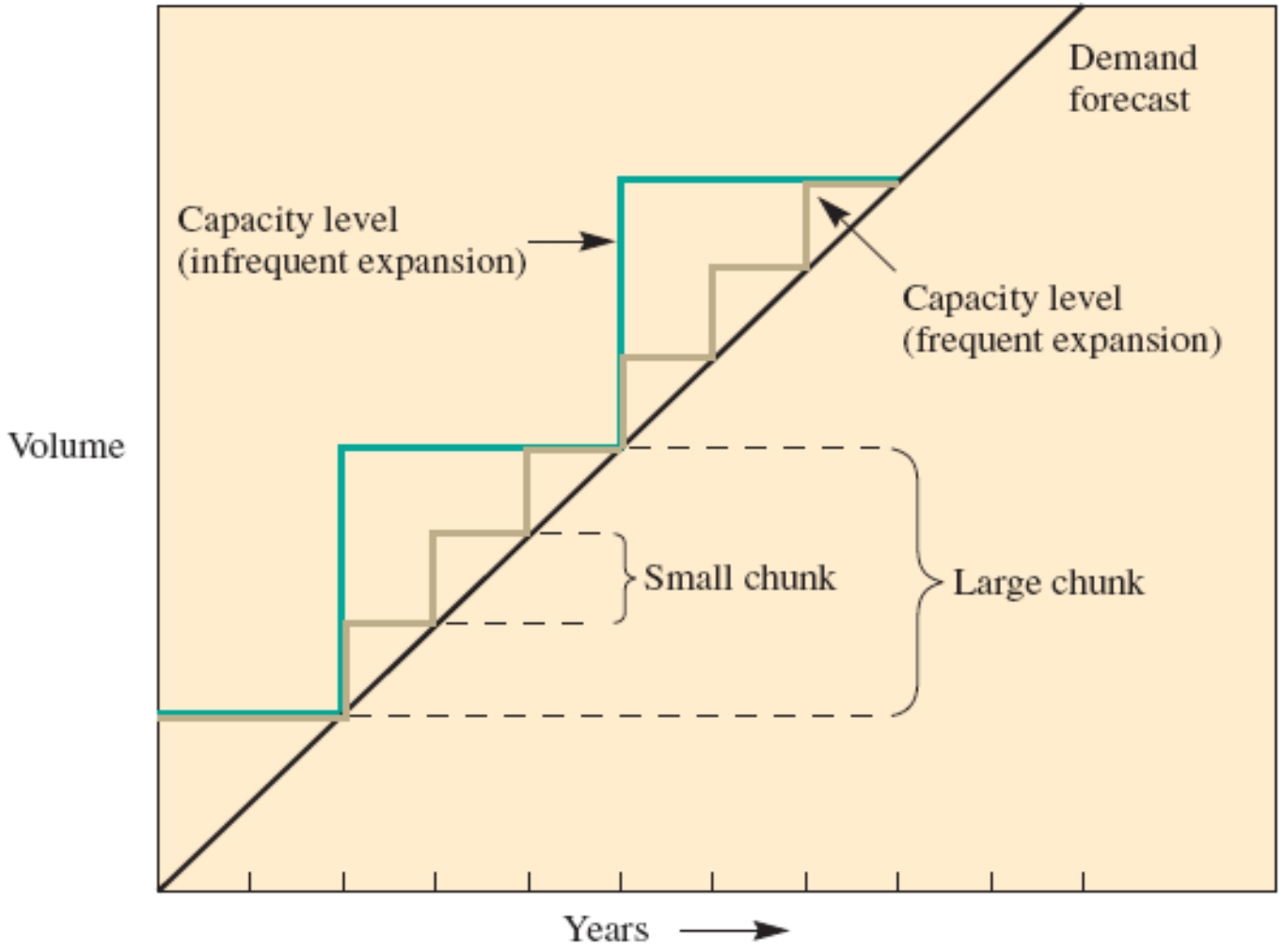
Applications: PCB/MB, memory, panel, foundry manufacturing

Scheme 1: Regression based on downstream consumer-electronics

Scheme 2: Time series considering cyclic patterns, trends, and etc.

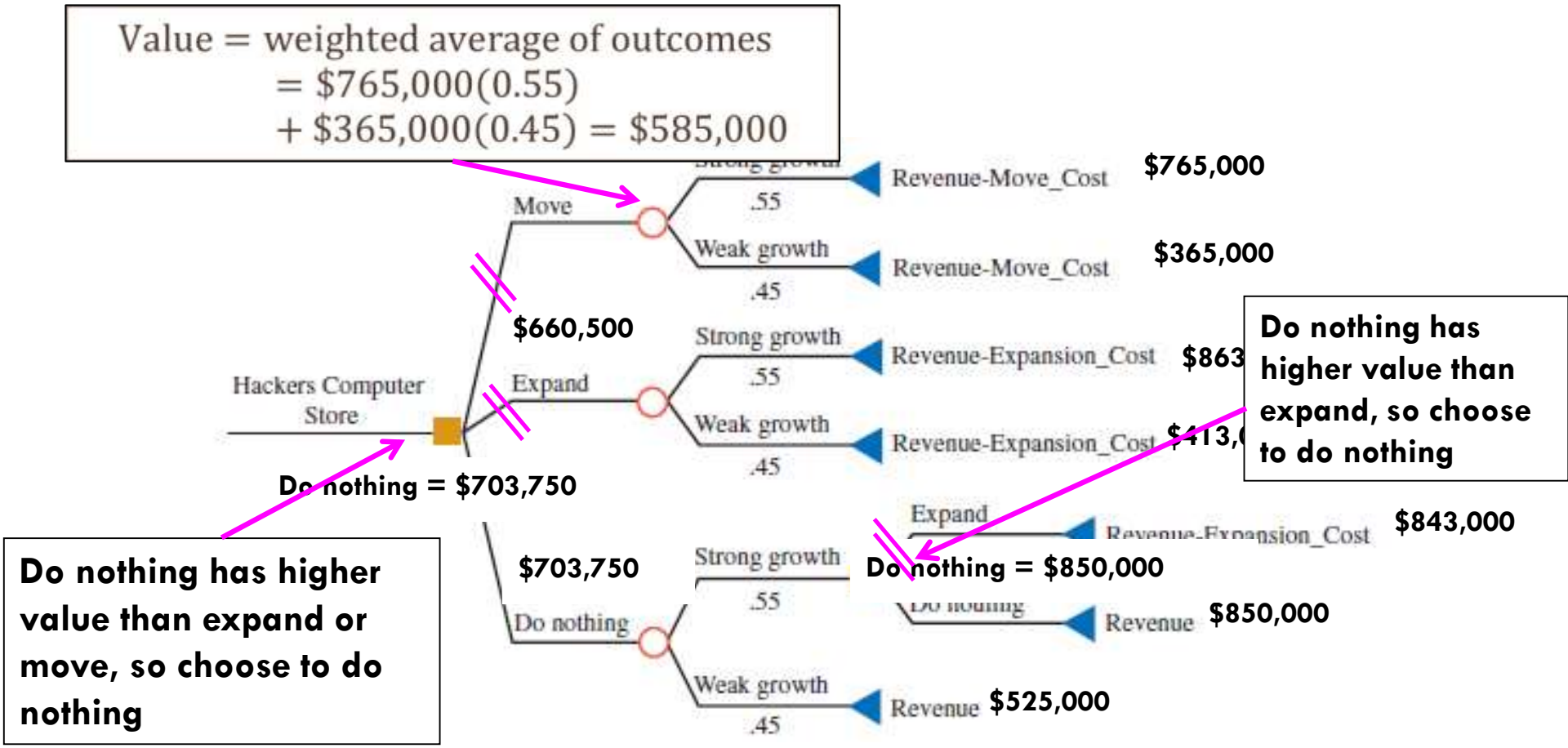
Scheme 3: Dynamic forecasting considering interactive dynamics (between the competing brands, firms, vendors, and etc.)

Capacity planning (產能規劃)

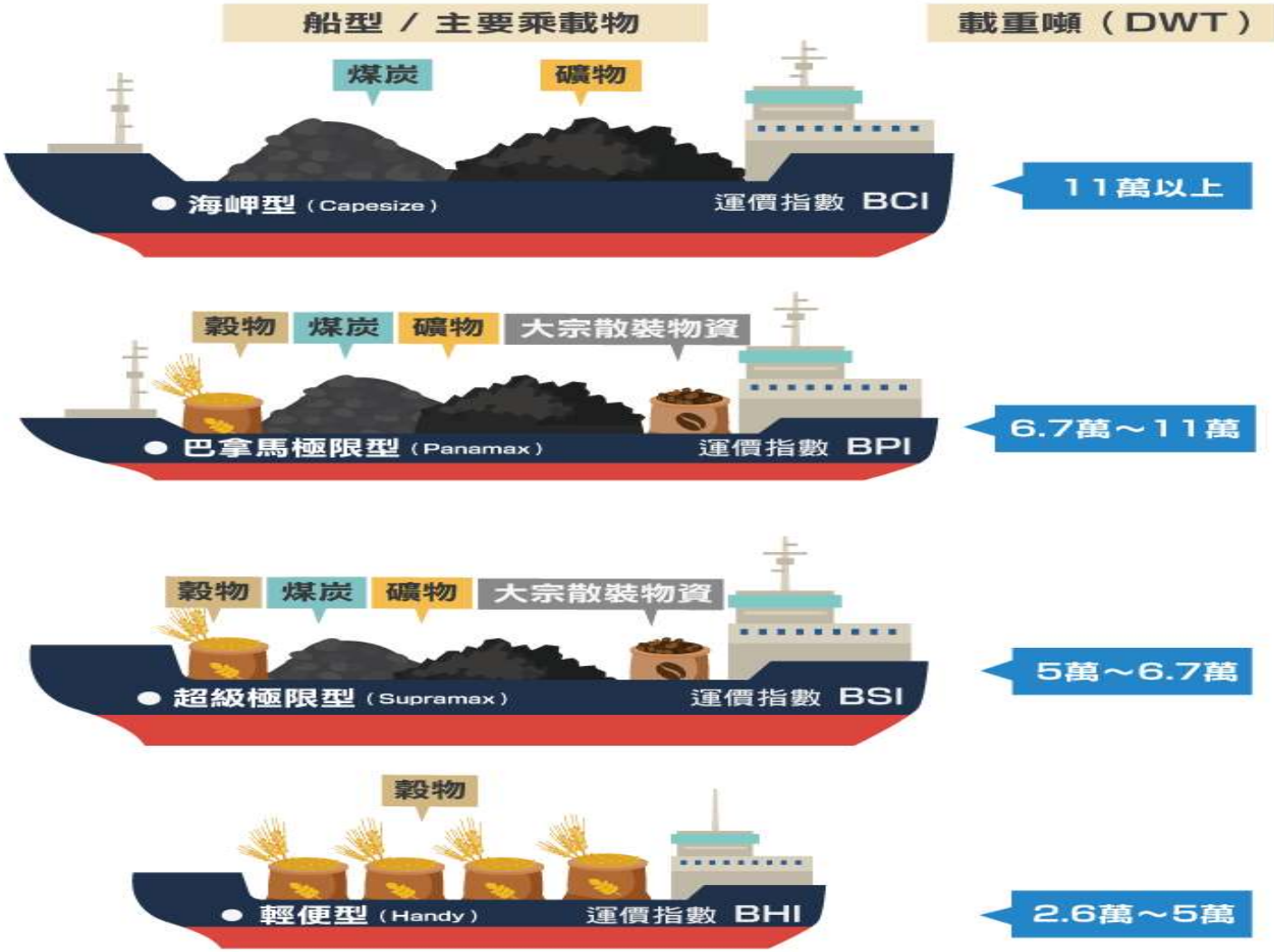


利用決策分析進行規劃為何常常失敗呢？

- Work backwards to calculate the value of each decision/event



Bulk_carriers



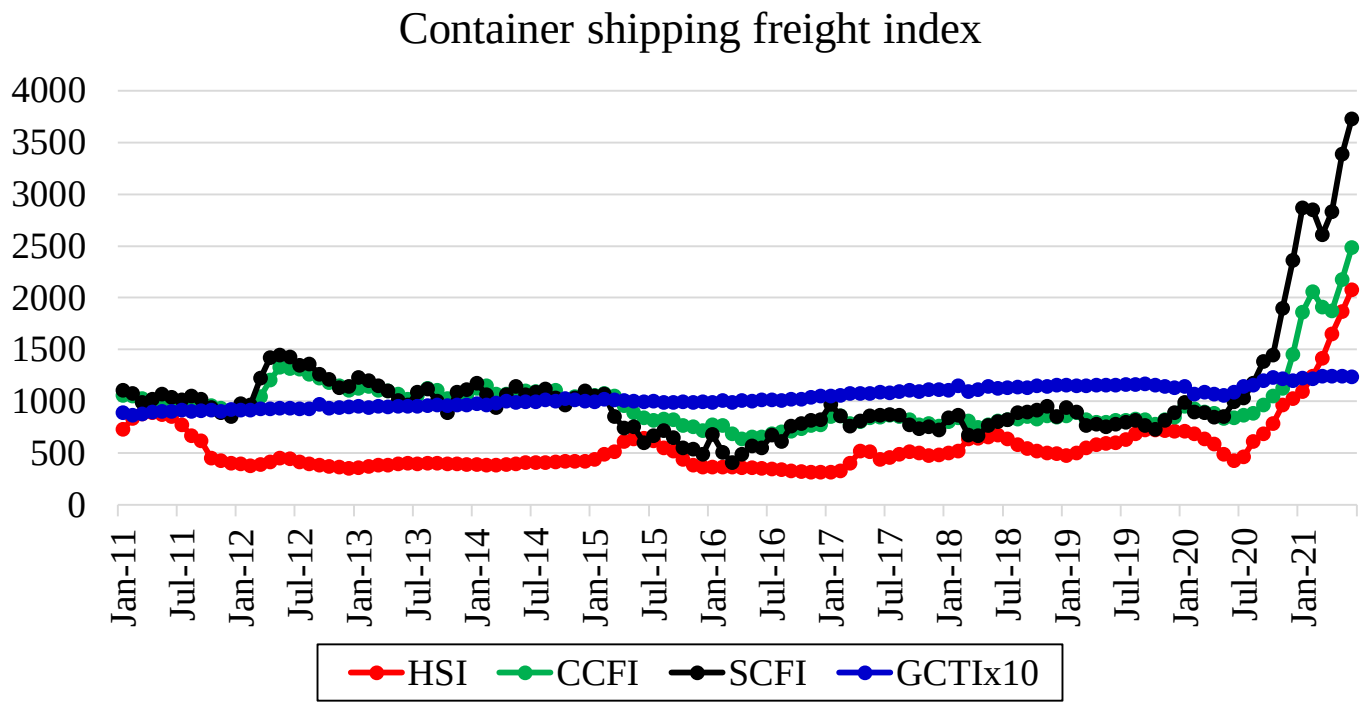
綜合比較

公司	營收占比（%）			
	美國線	歐洲線	亞洲線	其他
長榮	44.0	24.0	21.0	11.0
陽明	42.0	32.0	26.0	0
萬海	21.6	0	56.9	21.5

說明：資料為2020年報；陽明亞洲線含澳洲與南美線
資料來源：各業者

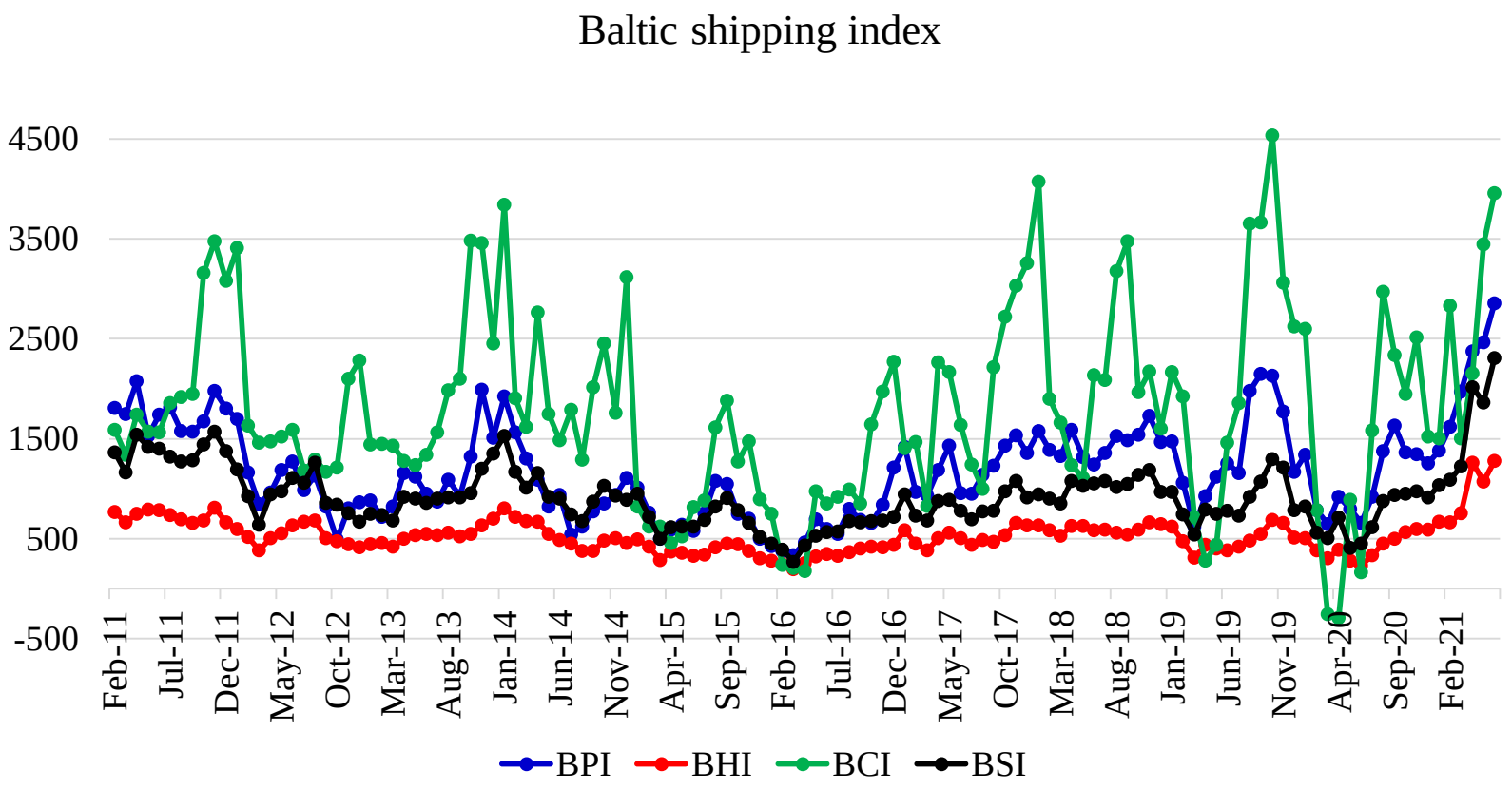
船型	慧洋-KY	四維航	台航	裕民	新興	中航
海岬型	3	1	0	15	9	10
巴拿馬極限型	29	6	6	11	2	0
超級極限型	28		9	6	2	0
輕便型	43	31	0	0	0	0
小型輕便型	32		0	0	0	0
其他船型	3	17	11	11	4	0
總船數	138	55	26	44	17	10

Maritime indices for container ships



GCTI (global container throughput index), HSI (Harpex shipping index), SCFI (Shanghai containerized freight index), CCFI (China containerized freight index)

Maritime indices for bulk carriers

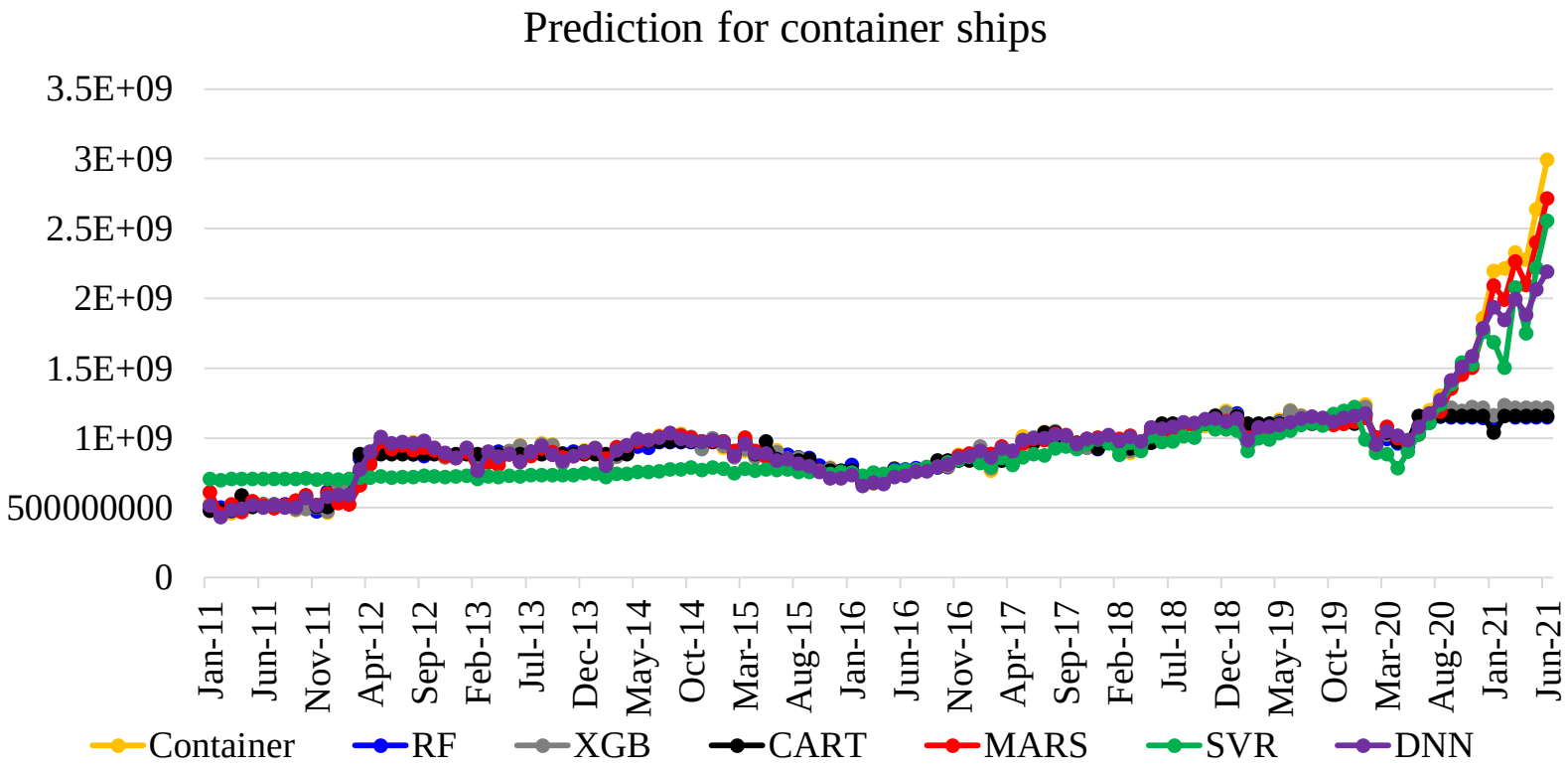


BCI (Baltic Capesize Index), BPI (Baltic Panamax Index), BSI (Baltic Supramax Index), BHI (Baltic Handysize Index)

Key performance indicators for container ships

Container	RF	XGB	CART	MARS	AVG
GCTI	18.27%	11.75%	23.40%	21%	18.61%
HSI	20.59%	4.77%	44.48%		17.46%
CCFI	17.98%	5.96%		29%	13.24%
SCFI	25.90%	10.39%	6.93%		10.81%
Retail sales index (European Unit)	11.10%	9.03%	17.90%		9.51%
IPI (Taiwan)	2.70%	7.84%	3.83%	21%	8.84%
Retail sales index (China)	0.21%	10.22%		14%	6.11%
Export of base metals	0.02%	4.26%		14%	4.57%
Import of chemicals	0.08%	3.75%		14%	4.46%
Export of chemicals	0.70%	7.84%			2.14%
Retail sales index (United States)	0.68%	6.98%			1.92%
Import of base metals	0.12%	4.09%	3.45%		1.92%
Seasonal factor	0.28%	4.60%			1.22%
Import of food goods	0.07%	4.09%			1.04%
Crude Oil	1.20%	1.70%			0.73%
Export of plastics and textile goods	0.10%	2.73%			0.71%

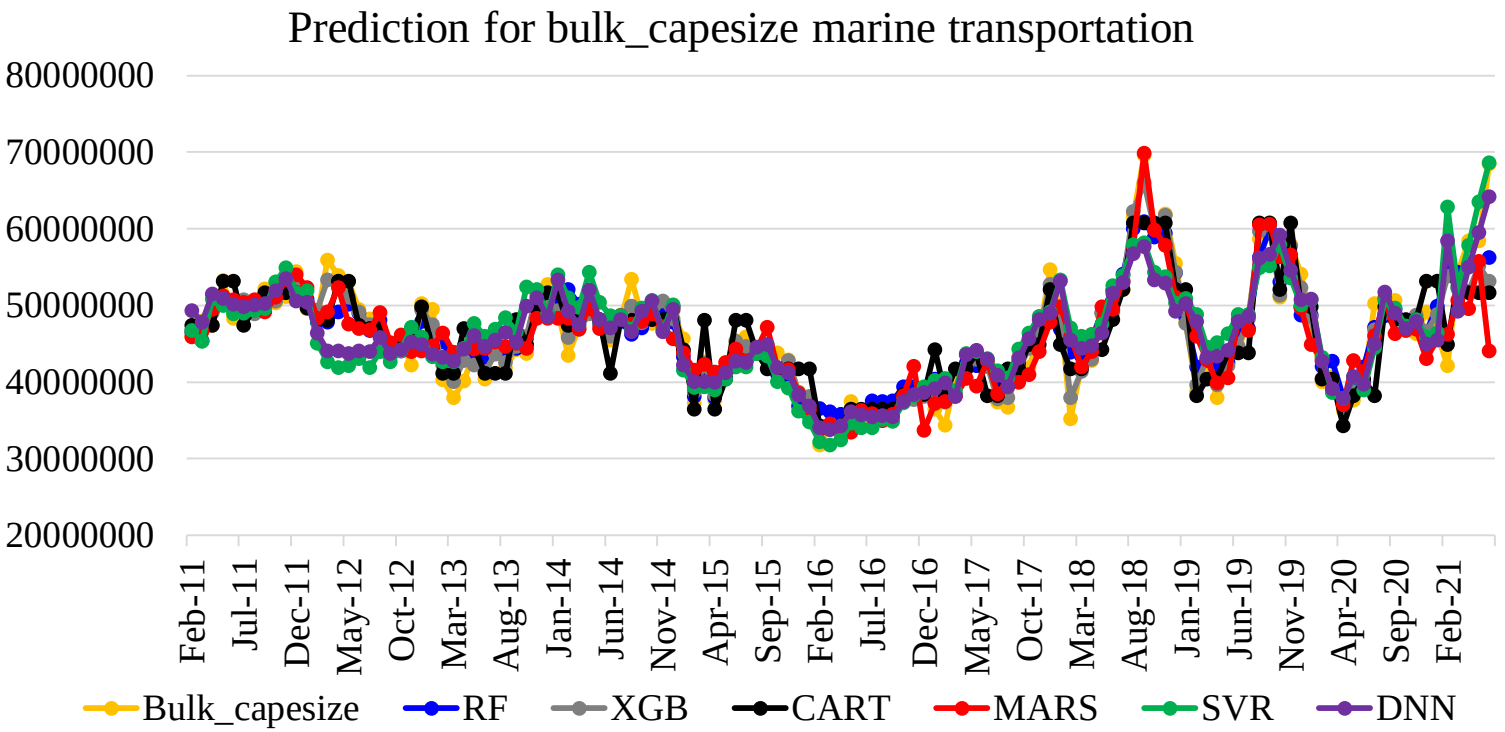
Sales forecasting for container ships



Key performance indicators for capesize carriers

Container	RF	XGB	CART	MARS	AVG
BSI_t-1	7.07%	4.78%	29.06%	17%	14.48%
PPI (United States)	25.20%	5.48%	13.34%	6%	12.51%
BCI_t-1	10.21%	11.94%	5.77%	14%	10.48%
BPI_t-1	24.78%	3.65%	5.77%	4%	9.55%
Australian thermal coal	4.36%	6.74%	6.29%	15%	8.10%
Soybean	3.43%	9.55%	12.72%	5%	7.68%
Wheat	3.11%	11.94%	3.59%	12%	7.66%
Corn	2.94%	6.74%	2.29%	9%	5.24%
Metal production (China)	1.76%	10.53%	5.66%	3%	5.24%
BHI_t-1	1.25%	4.35%	1.81%	10%	4.35%
Crude Oil	2.84%	5.76%	3.61%	5%	4.3%
Seasonal factor	4.79%	3.09%	4.37%		3.06%
Rock phosphate	1.55%	4.21%	2.44%		2.05%
Iron ore	1.18%	4.49%	2.04%		1.93%
Base metal index	2.81%	3.09%	1.75%		1.91%
PPI (China)	2.72%	3.65%			1.59%

Sales forecasting for capesize carriers



Key performance indicators for handysize carriers

Container	RF	XGB	CART	MARS	AVG
Metal production (China)	49.94%	14.80%	36.43%	55.00%	39.04%
Australian thermal coal	7.43%	16.47%	19.91%	17.00%	15.20%
Rock phosphate	4.55%	11.69%	5.37%	2.00%	5.90%
PPI (China)	1.85%	7.16%	6.24%	4.00%	4.81%
PPI (United States)	3.93%	7.88%	7.35%		4.79%
BHI_t-1	2.60%	6.44%	8.17%		4.30%
BPI_t-1	0.19%	8.35%	0.34%	8.00%	4.22%
Crude Oil	0.70%	5.49%	3.15%	6.00%	3.84%
Soybean	8.80%	3.10%	2.32%		3.56%
Iron Ore	8.57%	4.06%	0.00%		3.16%
Corn	2.68%	4.30%	1.39%	3.00%	2.84%
BSI_t-1	6.42%	2.15%	0.38%		2.24%
Base metal index	1.35%	0.95%	2.48%	4.00%	2.20%
Wheat	0.47%	2.63%	3.20%		1.58%
Seasonal factor	0.12%	3.10%	1.11%		1.08%
BCI_t-1	0.40%	1.43%	2.11%		0.99%

Sales forecasting for handysize carriers

